Cloud-Aware Learning-Based Attitude Control for Rainforest Earth Observation

Semester Project

MSc Mechanical Engineering, ETH Zürich

Cédric Grieder

supervised by Prof. Dr. Marco Hutter (ETH RSL); Alejandro Torrents Rufas & Mathias Burkhalter (Beyond Gravity)

Robotic Systems Lab, ETH Zürich | Semester project at Beyond Gravity

July 23, 2026

Abstract

Satellite imaging of tropical rainforests is limited by persistent cloud cover. Conventional controllers image every target on a fixed pointing schedule, regardless of cloud conditions. A limited *shutter budget* (a hard cap on captures per pass) and downlink capacity are wasted on cloud-obscured frames scientists cannot use. The intended remedy is onboard opportunity selection: a forward-looking fisheye reveals upcoming targets and clouds before they enter the main imaging window, so a policy can prefer high-value clear captures and skip occluded ones under that budget.

Because that idea appears straightforward, this work asks a narrower question: what reward shaping and action-space design are required to train such a policy in simulation? We build a reproducible platform for that investigation: single-axis attitude dynamics, ray-tracing cameras, parameterised cloud primitives, and a multi-phase safety controller. A deterministic sequential-target overflight policy serves as the comparison baseline. Two off-policy reinforcement-learning algorithms are trained and evaluated: Soft Actor-Critic and Maximum a Posteriori Policy Optimisation.

The experiments show that the problem is not trivial in practice. Reward structure, how the agent commands attitude, and correct algorithm implementation each had to be resolved before useful behaviour emerged. Both algorithms eventually develop selective shuttering absent in the deterministic baseline, but reliable mission-level gains over that baseline were not established within the semester scope.

The study is limited to a 2D single-pass orbital model. Attitude safety constraints are enforced throughout. Flight hardware, multi-orbit planning, and three-dimensional operations are out of scope.

Contents

Nomenclature

0.1 Abbreviations

Abbreviation	Definition
CNN	Convolutional neural network (vision observation encoder)
EO	Earth observation
FOV	Field of view
GSD	Ground sample distance
LLA	Latitude, longitude, altitude (geodetic coordinates)
LOS	Line of sight
MLP	Multilayer perceptron (scalar observation branch)
SAC	Soft Actor-Critic
MPO	Maximum *a posteriori* policy optimization
OBC	On-board computer
PD	Proportional–derivative (pointing controller)
RL	Reinforcement learning
RW	Reaction wheel
WGS84	World Geodetic System 1984 (reference ellipsoid)

0.2 Symbols

Symbol	Meaning
R_{E}	Mean Earth radius (orbital geometry)
μ	Earth gravitational parameter
h	Orbit altitude above mean Earth radius
I_{sat}	Spacecraft moment of inertia (2D model: I_{zz})
$\boldsymbol{\tau}$	Reaction-wheel torque command [N m] (scalar in the 2D kernel)
θ_{orb}	True anomaly of the satellite (= orbit-plane angle in circular orbit)
θ_{body}	Body z-axis angle in the orbit disk
ω_{sat}, ω_{w}	Spacecraft body rate and wheel speed
f_{cloud}	Primary-camera cloud-blocked fraction along the sensor strip

0.3 Physical and simulation constants

Unless noted, values are fixed inputs; h and cloud layouts are randomized per episode (Table 3).

Table 1: Earth, orbit, and spacecraft parameters.

Quantity	Symbol	Value	Unit / note
Mean Earth radius	R_E	6371.0088	km
Gravitational parameter	μ	398 600.4418	km^3/s^2
Geodetic model	—	WGS84 ellipsoid (surface / LOS)	
Orbit altitude (episode)	h	uniform on [510 km, 570 km]	
Satellite mass	m_{sat}	250	kg
Moment of inertia (3D)	I_{xx}, I_{yy}, I_{zz}	16.6, 21.7, 31.2 $kg\,m^2$	
Moment of inertia (2D sim.)	I_{sat}	31.2	$kg\,m^2$ (I_{zz})
RW max torque	τ_{max}	0.1	N m
RW max momentum	H_{max}	0.4	N m s
Star-tracker rate limit	—	3	°/s
Off-nadir hard limit	—	45	°

Table 2: Camera and simulation timing.

Quantity	Symbol	Value	Unit / note
Sensor resolution	—	9344 × 7000 pixels (cross × along)	
Pixel pitch	—	3.2	µm
Focal length	f	1067	mm
Exposure time	t_{exp}	100	µs
Simulation step	Δt	1.5	s
Max captures per orbit	—	10	shutter budget (resource proxy)
Secondary wide-angle (fisheye) camera			
Sensor resolution	—	5312 × 2988 pixels	
Pixel pitch	—	1.55	µm
Vertical FOV	—	70	°
Boresight tilt from nadir	—	25	°

Table 3: Mission profile and reward scales (default training configuration).

Quantity	Value / range
Target grid	50 areas, 15 km extent, 40 km spacing along corridor
Cloud patches per episode	uniform integer in [15, 30] along corridor
Cloud horizontal extent	1 km to 80 km
Cloud base altitude	4 km to 12 km
Cloud thickness	1 km to 16 km; top capped at 20 km
Capture reward weight	$w_{capture} = 100$
Optimal / threshold ground range	$d_{op} = 400$ km, $d_{th} = 1500$ km
Viewing-distance outer gate	2500 km

1 Introduction

1.1 Motivation

Climate and environmental science depend on repeat optical imaging of regions such as tropical rainforests, where cloud cover is frequent and unpredictable at the timescale of a satellite pass. Mission operators today often rely on deterministic strategies with fixed nadir attitude and pre-planned shutter times that treat every target along a ground track equally. When clouds occlude the line of sight, those strategies still consume onboard memory and downlink budget on low-value frames, while scientists lose observations that may not be recoverable on the next pass. The cost is therefore threefold: reduced science yield, wasted capture capacity, and unnecessary actuator activity from indiscriminate pointing and slewing.

Autonomous onboard decision-making is motivated by latency: ground-in-the-loop replanning cannot react to the cloud layout actually present during the pass. Both *when* to expose the shutter and *how* to orient the spacecraft for maximal image quality therefore matter, and they must respect attitude safety limits on reaction-wheel use and pointing envelopes. From the outside, the required behaviour looks straightforward: use onboard camera feedback to skip cloud-occluded targets and capture only when the view is clear. This report treats that apparent simplicity as the starting point and asks what training design is actually required to reach it.

The specific orbit geometry is not tied to a polar versus equatorial regime. Any circular orbit viewed at the instant of overflight is equivalent for the control problem posed here.

Current operational process. In a typical Earth-observation mission today, domain scientists compile a list of target areas they wish to image and submit it to mission control. The ground segment then builds an observation schedule: it checks which targets fall within the satellite's ground track on upcoming passes, verifies observable constraints such as solar illumination and off-nadir angle limits, and uplinks a fixed command sequence. This process works well for constraints that can be predicted from orbital mechanics. Cloud cover, however, cannot be scheduled out: the weather state at the moment of overflight is unknown at planning time, and the uplinked schedule cannot be modified during the pass. The satellite therefore executes its pre-planned capture sequence regardless of what the cameras actually see, consuming onboard storage, electrical power, downlink capacity, and attitude-control effort on frames that may be entirely cloud-occluded [3].

1.2 Intended solution concept

Today's operational process is ground-planned: the capture list is fixed before the pass, and uplink latency is too large for cloud-aware replanning during overflight. Under that paradigm a forward-looking sensor has little operational value, because the satellite cannot act on what it sees ahead. Consequently, current missions typically do not fly a dedicated lookahead imager for in-pass scheduling.

An onboard scheduling agent changes that premise. If capture decisions are made during the pass, the agent needs information about upcoming targets and cloud conditions *before* those scenes enter the primary imaging window. Otherwise it can react only when a target is already under the science boresight, which is too late for useful opportunity selection under a limited shutter budget.

The primary payload is a narrow-field science camera (hyperspectral-class in the mission concept). Pointing that instrument forward would still cover only a thin along-track strip, so it is a poor lookahead sensor for surveying multiple upcoming opportunities. This study therefore adds a second, forward-tilted wide-angle (fisheye) camera that observes the corridor ahead while the primary camera remains dedicated to science capture near nadir (Table 2; 70° along-track FOV, 25° prograde tilt). Figure 1 illustrates this sensing geometry on a representative along-track pass.

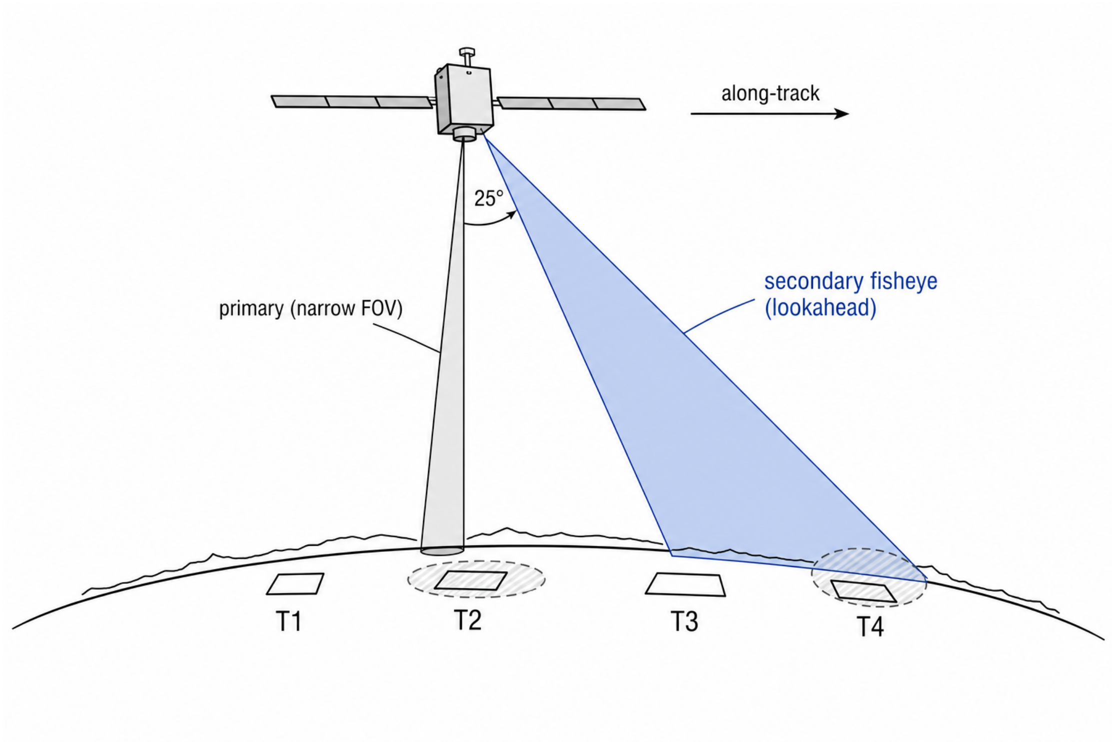


Figure 1: Lookahead sensing geometry. The narrow primary camera images near nadir while a forward-tilted fisheye observes upcoming targets along-track before they enter the primary footprint (25° prograde tilt; Table 2).

Lookahead sensing is not unique to a fisheye. Alternatives include cloud or weather products from other satellites, relayed via the ground segment or cross-link, and those products need not be image-shaped at all. We evaluate the onboard fisheye approach because it is deployment-feasible on a single spacecraft and avoids network effects and external data dependencies: as long as the satellite has power and a downlink path for the science data it produces, the scheduling loop can operate from its own sensors.

The intended decision process is sequential opportunity selection under scarcity. The fisheye reveals clear and cloud-occluded candidates ahead of the primary footprint; an onboard policy ranks those imaging opportunities and spends the limited shutter budget (proxying storage, downlink capacity, and actuator effort) on high-value clear captures rather than executing a fixed pre-planned schedule. This report then asks a narrower training question: which reward shaping and action-space designs make that selective behaviour learnable in simulation.

1.3 Research question

Cloud-aware autonomous scheduling is an obvious improvement over fixed pre-planned capture. The motivating question of this report is therefore not whether such a policy should exist, but how to train one in simulation:

> **What reward shaping and action-space design are sufficient for a reinforcement-learning agent to learn selective pointing and shuttering during a cloud-occluded Earth-observation pass?**

Capture decisions are sequential and uncertain. Cloud occlusion, attitude dynamics, and discrete shutter events interact over a single pass, so the design of the learning interface matters as much as the algorithm choice. The experiments below treat this as a design-space question: which training choices make selective shuttering learnable, and which do not.

1.4 Related work

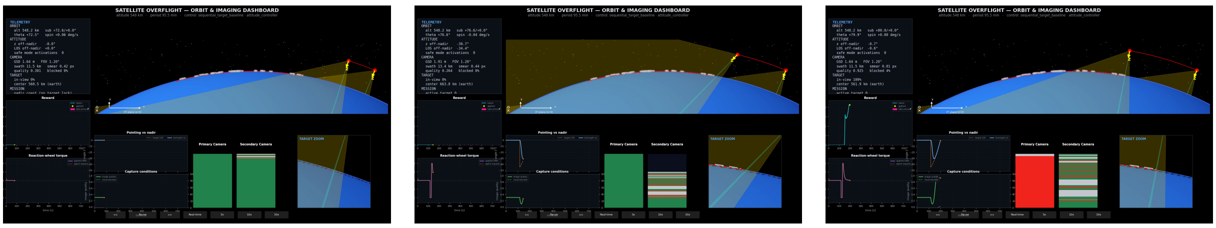
Earth-observation satellite scheduling is a well-studied operations-research domain. Surveys of satellite scheduling problems document how limited onboard storage, downlink, and energy constrain capture opportunities, and how waste of those resources reduces science yield [3]. Much of that literature addresses ground-segment planning over multi-orbit horizons. The present work instead asks how a single-pass, onboard policy should decide pointing and shuttering when cloud cover is revealed only during the overflight.

Continuous-control reinforcement learning provides candidate algorithms for that onboard layer. Soft Actor-Critic (SAC) combines off-policy sample reuse with a maximum-entropy objective, which tends to stabilise stochastic continuous control relative to brittle deterministic off-policy methods [5]. Maximum a Posteriori Policy Optimisation (MPO) frames improvement as relative-entropy-constrained policy iteration with separate trust regions on the policy mean and covariance [1]. Both algorithms are used in the experiments below as interchangeable learners over a shared environment and observation stack.

Beyond the policy update, observation structure matters when sensors are heterogeneous. Integrating state representation learning into deep RL shows that reward alone may not shape a usable embedding when modalities must be fused, and that structured encoders improve sample efficiency and generalisation [2]. For tabular and numeric inputs specifically, embedding numerical features before mixing outperforms feeding raw scalars through a shallow multilayer perceptron [4]; that motivation underlies encoder variants that compress structured bearing and mask fields before fusion with vision features.

Prior work therefore covers Earth-observation resource scheduling, stable continuous-control RL, and structured observation encoders. It does not, however, map which reward shaping and action-space designs are sufficient for cloud-aware shuttering under hard attitude-safety constraints in this simplified Earth-observation setting. That mapping is the focus of the present report.

1.5 Deterministic baseline



(a) Approach (b) Point (c) Capture

Figure 2: Deterministic baseline overflight (orbit-disk dashboard). Left to right: (a) Approach: nadir coast into the target corridor before the first lock; (b) Point: large off-nadir slew toward the scheduled target bearing; (c) Capture: return to nadir after the first shutter, with high image quality and the reward spike still visible. The schedule fires regardless of cloud cover (Table 5).

The reference strategy is a pre-scheduled sequential controller used as the comparison anchor throughout the experiments (Section 2.5): an OBC proportional–derivative loop commands target body angles (nadir coast or target bearing) under the same attitude-safety stack as the learned policy, with no cloud awareness and no adaptive capture list (see Table 5).

A real mission is additionally constrained by onboard memory, downlink bandwidth, and energy [3]. We model this resource limit with a *shutter budget*: a hard cap of ten capture actions per episode. Because the deterministic baseline treats every scheduled target equally, that budget is consumed indiscriminately across clear and cloud-occluded targets alike. This yields a fixed, predictable observation cadence but wastes capture capacity on frames that are scientifically unusable when a cloud is in the field of view.

1.6 Scope and simplifications

The model deliberately stays close to the control problem while omitting much operational detail: 2D attitude dynamics, basic circular-orbit geometry, physical ray tracing for cameras and clouds, safety arbitration, and single-pass episodes. It does not model multi-orbit scheduling, flight hardware, or real imagery ingestion. Illumination conditions are not modelled: all episodes are treated as daytime passes with perfect solar lighting. This is a deliberate simplification. Optical Earth observation is not performed at night, and solar geometry only excludes certain orbit geometries rather than driving frame-to-frame image quality during a pass. Cloud occlusion is a far stronger and more variable image quality constraint within a single pass, and is the focus of this work. Section 2 describes how the question was investigated and lists modelled subsystems explicitly; Discussion states extensions (three-dimensional attitude, richer terrain, stronger RL benchmarks) as logical next steps rather than deliverables of this semester project.

Long-term context. The intended onboard stack is hierarchical: a mission planner sets observation intent, a learned policy decides when and what to image, and an OBC executes maneuvers through PD pointing under hard attitude-safety limits. This report maps the learning interface for that scheduling layer, not the full flight stack. Section 5.3 details the control hierarchy and the torque-to-vector action-interface path explored in the experiments.

1.7 Contributions

This report documents:

- **Platform.** A simulation-and-learning pipeline with seeded mission randomisation and matched baseline comparison, so that reward and action-space variants are tested under identical cloud and target draws.
- **Design-space study.** A sequence of fourteen experiments that maps which reward terms, action representations, and algorithm choices enable cloud-aware shuttering behaviour, and which do not.
- **Evaluation protocol.** Training and evaluation conventions with a fixed environment seed, baseline-policy buffer seeding, and a deterministic baseline as the comparison anchor.
- **Evidence.** Qualitative and quantitative artefacts from frozen runs, including selective shuttering behaviour in late-training episodes where learning succeeded.

The engineering platform and the experimental map are the primary deliverables; mission-level outperformance of the baseline was not established within the semester scope (Section 5).

1.8 Report structure

Section 1.2 states the intended onboard sensing and scheduling concept. Section 1.4 situates the work against Earth-observation scheduling surveys and continuous-control reinforcement learning. Section 2 describes the research approach, simulation environment, and learning method. Section 3 documents the design-space search: which reward and action-space choices were tested and what each revealed. Section 4 aggregates the behavioural and return outcomes. Sections 5 and 6 interpret the mapped design space, rejected shortcuts, and outlook. Technical constants are listed in the Nomenclature section; source-code entry points are in the Appendix.

2 Methods

This section describes how the research question (Section 1.3) was investigated. We built a simulation platform and ran a sequence of controlled experiments that varied reward terms, action representation (direct torque versus nadir-relative pointing offset), and learning algorithm. Each configuration was evaluated against the deterministic baseline (Section 1.5) under matched environment draws: same altitude, cloud layout, and target positions per episode. Success is measured by whether the agent learns a stable policy that uses vision feedback to shutter selectively rather than on every step.

The subsections below document every modelling choice, constant, and algorithm in sufficient detail to reproduce the experiment results. Numeric values appear in the Nomenclature and in the tables below; repository source locations are listed in the Appendix.

2.1 Environment overview

The mission is a single-satellite nadir-looking Earth-observation overflight. One episode represents a single contiguous pass over a ground corridor.

Altitude and orbit. The simulation uses a 2D orbit-plane model: the satellite moves as a point on a circle whose true anomaly θ_{orbit} advances at the instantaneous circular orbit rate (in a circular orbit, true anomaly equals orbital angle). Orbital altitude is sampled uniformly from [510 km, 570 km] per episode; all other geometry scales with altitude, so the choice of specific ground track is arbitrary.

Target corridor. Fifty observation targets are placed at fixed positions along a ground corridor directly below the orbit plane. Their angular positions in the orbit plane are computed once at episode initialisation from the sampled altitude. The specific geographic location of the corridor is irrelevant: the agent only ever sees orbit-plane angles, cloud fractions, and attitude state.

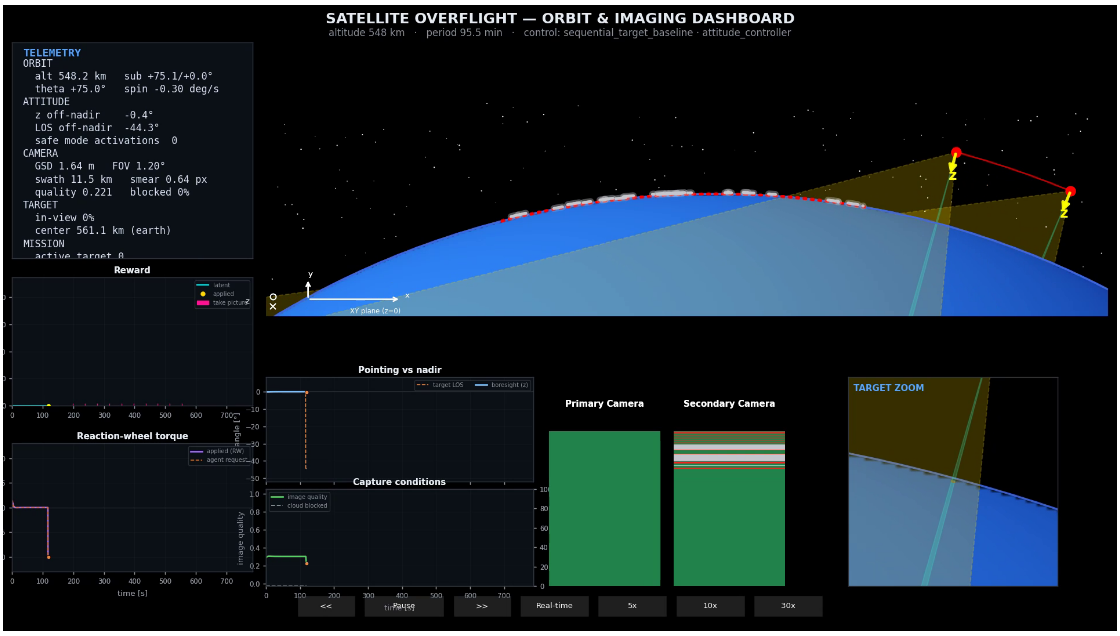


Figure 3: Environment overview on the orbit-disk dashboard early in the pass: satellite on the arc, target corridor markers on the Earth limb ahead of the boresight, before the first scheduled slew.

Episode window. The episode start and end positions along the pass were tuned so that the first target is not yet visible in the secondary wide-angle camera at episode open, with a small margin at both boundaries. This trims dead coast time at the corridor edges while keeping the engagement arc inside the pass. At the canonical simulation timestep of $\Delta t = 1.5$ s this yields approximately 516 controller steps per episode (≈ 774 s).

Off-policy learning and replay buffer. Training uses two off-policy algorithms, SAC (Section 2.8) and MPO (Section 2.9): the policy is improved from stored past experience, not only from the latest rollout. After each simulation step the transition (observation, action, reward, next observation) is appended to a *replay buffer*, a fixed-capacity store of recent steps. Gradient updates draw random mini-batches from this buffer, so a rare rewarded event (a successful capture) can contribute to many parameter updates. *Capture credit* (the reward applied when a shutter yields a useful first-time observation of a target) is sparse in this environment, so both the simulation timestep below and the warmup protocol in Section 3 affect whether the buffer contains enough useful transitions to learn from.

Simulation timestep selection. Choosing Δt requires balancing three competing constraints:

1. **Controller authority:** Δt must be small enough that the low-level attitude controller can still respond to the reaction-wheel dynamics within one step.
2. **Agent decisiveness:** the agent must be able to take meaningful pointing and shutter actions on a timescale relevant to the target encounter windows.
3. **Reward signal fidelity:** capture credit arrives only at shutter events on clear targets, so most steps carry little or no reward. If Δt is too large, target encounters fall between steps and fewer rewarded transitions are stored per episode, leaving the replay buffer (Section 2.1) with too little signal for off-policy learning.

To find the largest Δt satisfying these constraints we ran a timestep parity sweep over four candidates: $\Delta t \in \{0.4, 0.8, 1.0, 1.5\}$ s, with controller update interval matched to Δt. Each candidate ran one baseline warmup episode and had to satisfy: at least one shutter and episode return $\geq 50\%$ of the 0.4 s reference. All four candidates passed. $\Delta t = 1.5$ s was selected as the largest parity-passing timestep, cutting the episode from ~ 1935 to ~ 516 controller steps and proportionally increasing training throughput.

2.2 Simplifications

Table 4 summarises which subsystems are modelled and which are deliberately omitted.

Table 4: Modelled vs. not-modelled subsystems.

Subsystem	Modelled	Not modelled
Orbit propagation	Circular 2D (disk model)	J_2, atmospheric drag, eclipse
Attitude dynamics	2D rigid body, reaction-wheel torque chain	3-axis, flexibility, RW friction
Camera	1D ray tracing, image quality (motion smear)	Radiometric calibration, TDI, MTF
Cloud occlusion	Parameterised cloud-arc primitives	Spatially correlated fields, multi-layer
Terrain	Spherical Earth (WGS84 for geodesy only)	DEM, shadow, surface reflectance
Multi-orbit mission	Single pass per episode	Revisit scheduling, downlink windows
Hardware latency	Not modelled	Onboard CPU budget, comms delay

The 2D simplification is the most consequential: all attitude variables are scalar (θ_{body}, ω_{sat}, ω_{wheel}), and out-of-plane dynamics are absent. This is consistent with the scope stated in the introduction and is a known limitation (Section 5.5).

2.3 Dynamics model

Equations of motion. The satellite body angle θ_b and inertial rate ω_{sat} evolve under a single-axis reaction-wheel coupling:

$$\dot{\omega}_{sat} = -\frac{\tau_{wheel}}{I_{sat}}, \qquad \dot{\omega}_{wheel} = \frac{\tau_{wheel}}{I_{wheel}}, \qquad \dot{\theta}_b = \omega_{sat}, \tag{1}$$

where $I_{sat} = 31.2\,\mathrm{kg\,m^2}$ (Table 1) and the wheel inertia $I_{wheel} = \tau_{max,RW}/\omega_{w,max} = 0.1\,\mathrm{N\,m}/150\,\mathrm{rad\,s^{-1}}$. Both are integrated forward with forward-Euler at $\Delta t_{sim} = 1.5\,\mathrm{s}$ (Table 2).

Reaction-wheel limits. The wheel applies a torque $|\tau_{wheel}| \leq \tau_{max,RW} = 0.1\,\mathrm{N\,m}$ and stores angular momentum up to $H_{max} = 0.4\,\mathrm{N\,m\,s}$. A low-level rate gate blocks commands that would accelerate the satellite body beyond the star-tracker rate limit of $\dot{\theta}_{max} = 3\,^\circ\,\mathrm{s^{-1}}$; decelerating torques are always passed.

Satellite mass. Bus mass is $m_{sat} = 250\,\mathrm{kg}$ (relevant only for orbit-speed computation; inertia is the primary dynamics parameter).

2.4 Overflight and observation model

Camera geometry. The primary nadir-looking camera parameters are listed in Table 2. The vertical field of view and ground-sampling distance (GSD) are computed analytically from sampled altitude using a pinhole model. At 540 km the nominal GSD is approximately 1.6 m.

Lookahead sensing. The secondary camera is a forward-tilted wide-angle (fisheye) imager: 25° off nadir in the prograde direction with a 70° along-track field of view (Table 2). It provides early visibility of upcoming targets and cloud occlusion so the controller can decide among multiple imaging opportunities before they enter the primary nadir footprint (Section 1.2).

Camera observation lines. At each simulation timestep the primary and secondary cameras are evaluated as one-dimensional observation lines spanning their vertical fields of view (Table 2). Each line reports target coverage and cloud occlusion along the boresight strip; both use the same ray-intersection kernel.

Image quality and motion smear. Capturing a useful image requires the boresight ground footprint to remain nearly stationary during the exposure window. The full derivation proceeds in three steps: (i) compute the boresight ground velocity v_{bore} from orbit and body rates; (ii) convert to a physical ground blur δ during exposure; (iii) map δ to a scalar quality score $q \in [0, 1]$.

Step 1: Boresight ground speed. In the 2D orbit-disk model the satellite position is

$$\vec{s} = r_{orb}\,[\cos\theta_{orb},\, \sin\theta_{orb}]^\top, \tag{2}$$

with orbital radius r_{orb} and true anomaly $\theta_{orb}(t)$ advancing at angular rate $\dot{\theta}_{orb} = \omega_{orb}$. The boresight unit vector in the disk plane is

$$\hat{d} = [\cos\phi,\, \sin\phi]^\top, \qquad \dot{\phi} = \omega_{body}, \tag{3}$$

where ϕ is the absolute boresight angle (body angle plus nadir direction) and $\omega_{body} = \dot{\theta}_{body}$ is the current body rate.

The boresight ray $\vec{s} + t\,\hat{d}$ intersects the WGS84 oblate spheroid ($a = 6\,378\,137$ m, $b = 6\,356\,752$ m) at slant distance $t^\star$ from the positive root of

$$A\,t^2 + B\,t + C = 0, \tag{4}$$

where

$$A = \frac{d_x^2 + d_y^2}{a^2} + \frac{d_z^2}{b^2}, \tag{5}$$

$$B = 2\left(\frac{s_x d_x + s_y d_y}{a^2} + \frac{s_z d_z}{b^2}\right), \tag{6}$$

$$C = \frac{s_x^2 + s_y^2}{a^2} + \frac{s_z^2}{b^2} - 1. \tag{7}$$

(All 3-D coordinates are obtained by embedding the 2D disk in the (x, z) ECEF plane.)

The boresight ground hit is $\vec{g} = \vec{s} + t^\star\,\hat{d}$. Differentiating implicitly with respect to time:

$$\dot{t}^\star = -\frac{\dot{A}\,(t^\star)^2 + \dot{B}\,t^\star + \dot{C}}{2A\,t^\star + B}, \tag{8}$$

where $\dot{A}$, $\dot{B}$, $\dot{C}$ are computed from $\dot{\vec{s}}$ (satellite velocity) and $\dot{\hat{d}}$ (boresight rotation rate):

$$\dot{\vec{s}} = r_{\mathrm{orb}}\,\omega_{\mathrm{orb}}\,[-\sin\theta_{\mathrm{orb}},\, 0,\, \cos\theta_{\mathrm{orb}}]^\top, \qquad \dot{\hat{d}} = \omega_{\mathrm{body}}\,[-\sin\phi,\, 0,\, \cos\phi]^\top. \tag{9}$$

The hit velocity is then

$$\dot{\vec{g}} = \dot{\vec{s}} + \dot{t}^\star\,\hat{d} + t^\star\,\dot{\hat{d}}. \tag{10}$$

The boresight ground speed is $v_{\mathrm{bore}} = \|\dot{\vec{g}}\|$.

Step 2: Ground blur during exposure. During a finite exposure $t_{\mathrm{exp}} = 100\,\mu$s the boresight footprint moves

$$\delta = v_{\mathrm{bore}}\,t_{\mathrm{exp}} \quad [\mathrm{m}]. \tag{11}$$

The pixel smear in GSD units is

$$\sigma = \frac{\delta}{\mathrm{GSD}_{\mathrm{eff}}}, \tag{12}$$

where the effective GSD at off-nadir angle α is $\mathrm{GSD}_{\mathrm{eff}} = \mathrm{GSD}_0/\cos\alpha$, with nadir GSD

$$\mathrm{GSD}_0 = \frac{p \cdot h}{f} = \frac{3.2\,\mu\mathrm{m} \cdot h}{1067\,\mathrm{mm}} \tag{13}$$

(pixel pitch p, altitude h, focal length f). At $h = 540$ km: $\mathrm{GSD}_0 \approx 1.62$ m, giving a nadir swath of 9344×1.62 m ≈ 15.1 km along-track.

Step 3: Normalised quality score. Rather than using smear in pixels (which depends on altitude), quality is expressed in physical ground units via a Lorentzian:

$$q = \frac{r}{\delta + r}, \qquad r = 0.30\,\mathrm{m}, \quad t_{\mathrm{exp}} = 100\,\mu\mathrm{s}. \tag{14}$$

Properties: $q \to 1$ as $\delta \to 0$ (perfect tracking); $q = 0.5$ at $\delta = r = 0.30$ m; q is indifferent to GSD (quality degrades from optical smear, not pixel count).

Characteristic regimes. At the canonical orbit ($h = 540\,\mathrm{km}$, $\omega_{\mathrm{orb}} \approx 1.08 \times 10^{-3}\,\mathrm{rad\,s^{-1}}$):

- **Nadir, zero body rate:** $v_{\mathrm{bore}} = r_{\mathrm{orb}}\,\omega_{\mathrm{orb}} \approx 7.6\,\mathrm{km\,s^{-1}}$, $\delta \approx 0.76\,\mathrm{m}$, $q \approx 0.28$.
- **Target-tracking ($\omega_{\mathbf{body}} = \omega_{\mathbf{orb}}$):** $v_{\mathrm{bore}} \approx 0$, $\delta \approx 0$, $q \approx 1.0$.
- **Half-tracking:** $\omega_{\mathrm{body}} = 0.5\,\omega_{\mathrm{orb}}$, $\delta \approx 0.38\,\mathrm{m}$, $q \approx 0.44$.

The quality factor directly enters the capture-credit reward (Eq. 16), giving the agent an incentive to slow the boresight ground motion by pointing toward and tracking each target across its observation window.

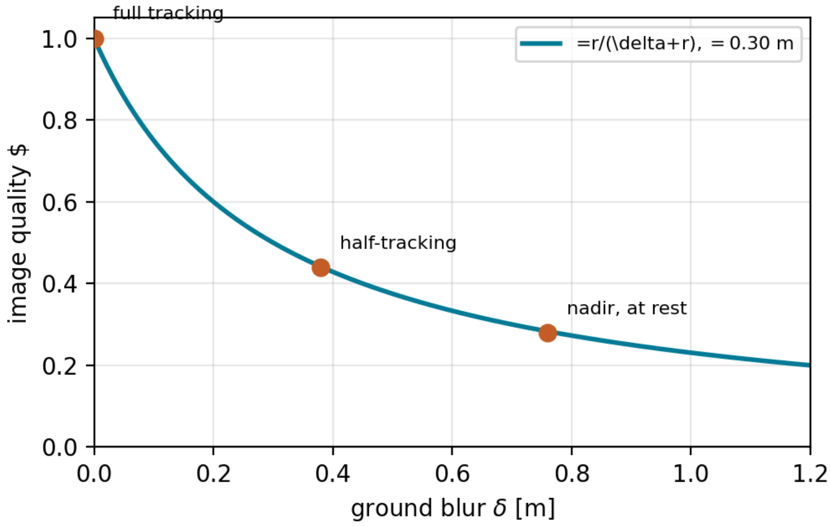


Figure 4: Normalised image quality $q = r/(\delta + r)$ with $r = 0.30\,\mathrm{m}$. Annotated operating points match the nadir-at-rest, half-tracking, and full-tracking regimes above.

Cloud model. Cloud primitives are parameterised arcs in the orbit plane. For each episode a seeded generator places cloud patches along the target corridor with extent, altitude, and thickness drawn from the ranges in Table 3. Over a single pass the clouds are effectively fixed relative to the ground corridor (any advective drift is negligible on the episode timescale); occlusion changes mainly because the satellite and camera rays move through that layout. The cloud-blocked fraction along each camera observation line is computed at every timestep and contributed to the per-step observation and reward.

2.5 Baseline strategy

The reference controller is a deterministic sequential-target overflight policy. Figure 2 shows the approach, point, and capture phases of a typical pass; Table 5 lists the scheduling logic, OBC PD gains, lock threshold, lead margin, shutter commands, and cloud awareness. The baseline uses the same attitude-safety stack and simulation kernel as the learned policy: safety overrides are not disabled.

In each learning run the same baseline policy fills five warmup episodes under that run's action interface and reward stack; the mean warmup return is therefore the matched *average baseline return* against which training and evaluation are compared (Section 3.15).

2.6 Attitude safety controller

Every torque command, whether from the learned policy or the baseline, passes through the attitude safety controller before reaching the reaction-wheel plant. The controller operates in two regimes:

Table 5: Deterministic baseline configuration (comparison anchor).

Aspect	Setting
Scheduling logic	Strided target list: 10 captures per pass from a 50-target corridor (indices 0, 5, 10, . . . , 45 on the evaluation seed). Nadir coast between engages; slew to target bearing when the orbit-angle window opens.
OBC pointing	PD loop on body-angle tracking error; gains $k_p \approx 1.15\,\mathrm{N\,m/rad}$, $k_d \approx 12.0\,\mathrm{N\,m\,s/rad}$ derived from $\tau_{\max} = 0.1\,\mathrm{N\,m}$, $I_{\mathrm{sat}} = 31.2\,\mathrm{kg\,m^2}$, and a 5° deceleration band.
Boresight lock threshold	15° between body boresight and target bearing (shutter fires once locked, or if image quality ≥ 0.25 while the target is in the sensor strip).
Lead margin	20° orbit-angle margin before target engage window opens.
Shutter commands	10 per episode (one per scheduled target; matches the episode capture cap).
Cloud awareness	None (captures on schedule regardless of occlusion).

Normal-mode taper. In nominal flight the controller applies a linear taper on the commanded torque when the satellite approaches the off-nadir hard limit:

$$\tau_{\mathrm{applied}} = \alpha\,\tau_{\mathrm{cmd}}, \qquad \alpha = \mathrm{clip}\left(\frac{\theta_{\mathrm{hard}} - \theta_{\text{off-nadir}}}{\theta_{\mathrm{hard}} - \theta_{\mathrm{arm}}},\ 0,\ 1\right), \tag{15}$$

where $\theta_{\mathrm{hard}} = 45°$ is the hard limit and θ_{arm} is the kinematic braking distance ($\theta_{\mathrm{arm}} = \theta_{\mathrm{hard}} - \omega^2/(2\tau_{\max}/I_{\mathrm{sat}})$).

Safe-mode entry condition. Safe mode is entered when the off-nadir angle equals or exceeds θ_{hard}, or when the predicted overshoot satisfies $\theta_{\text{off-nadir}} + \theta_{\mathrm{brake}} \geq \theta_{\mathrm{hard}}$.

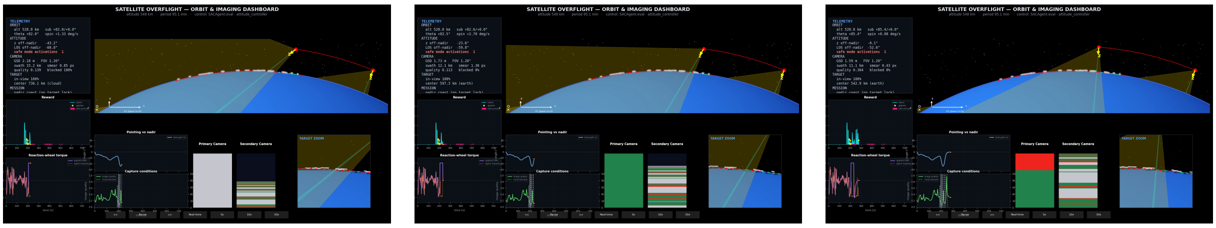
Safe-mode recovery phases. Once safe mode is active the agent torque command is replaced by an onboard recovery sequence (agent commands are ignored throughout):

1. **BRAKE:** zero-rate command until $|\omega_{\mathrm{sat}} - \omega_{\mathrm{orbit}}| \leq 0.1\,°\,\mathrm{s}^{-1}$.
2. **CRUISE:** slew toward nadir at $\pm 0.35\,°\,\mathrm{s}^{-1}$ until off-nadir $\leq 5.0°$.
3. **SETTLE:** zero-rate command until off-nadir $\leq 1.0°$ and rate $\leq 0.1\,°\,\mathrm{s}^{-1}$.
4. **LOCKOUT:** hold nadir for 60 s; agent torque rejected.

After LOCKOUT the controller exits safe mode and normal operation resumes. All phase thresholds are listed in Table 6.

Table 6: Attitude safety controller thresholds.

Parameter	Symbol	Value
Off-nadir hard limit	θ_{hard}	45°
Nadir recovery tolerance	θ_{tol}	1.0°
Decel start (CRUISE→SETTLE)	θ_{decel}	5.0°
Rate stop threshold	ω_{stop}	$0.1\,°\,\mathrm{s}^{-1}$
Safe-mode cruise rate	$\dot{\theta}_{\mathrm{crs}}$	$0.35\,°\,\mathrm{s}^{-1}$
Post-recovery lockout	t_{lock}	60 s



(a) High-torque slew (BRAKE) (b) CRUISE (c) LOCKOUT

Figure 5: One attitude-safety cycle in torque-mode evaluation (same episode; panels use the recovery-phase names BRAKE / CRUISE / LOCKOUT). (a) High-torque slew: the agent commands large torque and reaches $\approx 43^\circ$ off-nadir; safe mode activates (BRAKE) and rejects the request (applied torque drops while the agent request remains high). (b) CRUISE: onboard recovery slews the boresight back toward nadir (here $\approx 24^\circ$ off-nadir) while agent torque stays ignored. (c) LOCKOUT: nadir hold with agent torque rejected for 60 s. SETTLE (between CRUISE and LOCKOUT) is short and not shown.

2.7 Reward design

The reward is computed per controller step from the full episode simulation state. All active terms are listed below; inactive terms (distance reward, cloud penalty, energy penalty) were tested in early experiments and disabled by default (see Section 3).

Capture quality term (applied at shutter events). When the agent fires the shutter, is within budget, and the target has not been captured before:

$$r_{\text{cap}} = W_{\text{cap}} \cdot \text{cov} \cdot q \cdot (1 - f_{\text{cloud}}), \tag{16}$$

where $\text{cov} \in [0, 1]$ is the fraction of the primary camera footprint over the dominant target, $q \in [0, 1]$ is the image quality (Section 2.4), $f_{\text{cloud}} \in [0, 1]$ is the cloud-blocked fraction, and W_{cap} is the capture weight. Repeat shutters on an already-captured target consume budget but yield zero applied reward.

Budget-exhausted penalty. When the agent commands a shutter after the 10-capture budget is exhausted: $r_{\text{budget}} = -k_{\text{waste}}$ with $k_{\text{waste}} = 5.0$ (Exp 7, promoted to default).

Shutter-waste penalty (off by default). Optional: when applied capture credit $< \epsilon = 0.001$ on an accepted shutter: $r_{\text{waste}} = -k_{\text{waste}}$. Removed in Exp 9 (see Section 3.9).

Torque-effort regulariser (torque mode only).

$$r_{\text{effort}} = -k_{\text{eff}} \left(\frac{\tau_{\text{cmd}}}{\tau_{\text{max}}} \right)^2, \qquad k_{\text{eff}} = 0.1. \tag{17}$$

Applied every step on the agent-requested torque (before safety arbitration); disabled automatically in vector mode.

Safe-mode penalty (Exp 10 only, not default). Per-step penalty while the agent is in safe-mode recovery: $r_{\text{safe}} = -k_{\text{safe}}$, $k_{\text{safe}} = 2.0$. Ruled out at this coefficient (see Section 3.10).

2.8 Learning algorithm: SAC

We use Soft Actor-Critic (SAC) [5]. SAC is an off-policy actor-critic algorithm in the *maximum-entropy* reinforcement-learning framework: the stochastic policy maximises expected return while also maximising action entropy, so that high-reward behaviour is learned without collapsing prematurely to a near-deterministic action. In continuous control, this yields a stable off-policy learner that reuses replay-buffer experience and explores via the entropy term rather than via an additive noise process.

Soft critic update (policy evaluation). Two critics $Q_{\theta_1}, Q_{\theta_2}$ are trained against a shared soft Bellman target built from the minimum of the target critics (clipped double-Q), with an entropy bonus on the next action:

$$y = r + \gamma(1-d)\left(\min_{i\in\{1,2\}} Q_{\theta_i'}(s', a') - \alpha \log \pi_\phi(a'|s')\right), \tag{18}$$

where $a' \sim \pi_\phi(\cdot|s')$, $d \in \{0,1\}$ marks episode termination, and target networks $Q_{\theta_i'}$ are soft-updated with coefficient τ (Table 7). Each critic is fitted by mean-squared error to y.

Policy update (soft actor). The Gaussian actor π_ϕ is updated to maximise soft Q while retaining entropy:

$$J_\pi(\phi) = \mathbb{E}_{s\sim\mathcal{B},\, a\sim\pi_\phi}\Big[\alpha \log \pi_\phi(a|s) - \min_{i\in\{1,2\}} Q_{\theta_i}(s,a)\Big], \tag{19}$$

which is minimised by stochastic gradient descent (reparameterised samples; tanh-squashed actions with the standard change-of-variables correction on $\log \pi$).

Fixed entropy temperature. The temperature α trades off return against entropy. We use a fixed coefficient $\alpha = 0.2$ (no automatic temperature dual), matching the overnight SAC fork used throughout the experiment series. This is simpler than the learned-α variant of SAC and was sufficient for the sparse-reward regimes in which SAC first showed a learning signal (Exp 3).

Network architecture. Actor and critics use the same 1D-CNN encoder and fully connected widths as MPO (Section 2.9): two convolutional layers (kernel size 5, channels [16, 32], stride 2, embedding dimension 32), then actor head 90 units and critic heads 140 units. Hyperparameters are listed in Table 7.

Table 7: SAC hyperparameters (default configuration; all SAC experiments unless noted).

Parameter	Symbol	Value
Batch size	B	256
Replay buffer	$\lvert\mathcal{B}\rvert$	50 000
Discount factor	γ	0.99
Target network	τ	0.005
Actor learning rate	η_π	1.5×10^{-4}
Critic learning rate	η_Q	4.5×10^{-4}
Entropy temperature	α	0.2 (fixed)
Warmup episodes	—	5
Training episodes	—	50 (Exp 2: 7)
Eval episodes	—	2
Environment seed	—	7

2.9 Learning algorithm: MPO

We use Maximum a Posteriori Policy Optimisation (MPO) [1]. MPO frames policy improvement as a constrained relative-entropy problem solved by alternating E-step and M-step:

E-step (sample-based policy evaluation). Q^π targets are regressed from a target critic; the new non-parametric policy distribution $q^\star(a|s) \propto \pi(a|s)\exp(Q^\pi(s,a)/\eta)$ is derived by solving the dual:

$$\eta^\star = \arg\min_{\eta>0}\ \eta\,\varepsilon_\eta + \eta\log\mathbb{E}_\pi\left[\exp\left(\frac{Q^\pi(s,a)}{\eta}\right)\right], \tag{20}$$

where $\varepsilon_\eta = 0.1$ is the KL constraint on the temperature dual.

M-step (policy update with trust region). A parametric Gaussian policy π_θ is fitted to $q^\star$ subject to decoupled KL constraints on the mean and variance:

$$\max_\theta\ \mathbb{E}_{q^\star}[\log\pi_\theta(a|s)] \quad \text{s.t.} \quad \mathrm{KL}(\bar{\pi}_\mu \| \pi_{\theta,\mu}) \le \varepsilon_\mu,\ \mathrm{KL}(\bar{\pi}_\sigma \| \pi_{\theta,\sigma}) \le \varepsilon_\sigma, \tag{21}$$

with Lagrange multipliers $\alpha_\mu, \alpha_\sigma$ updated by dual ascent ($\varepsilon_\mu = 0.1$, $\varepsilon_\sigma = 0.01$; Table 8).

Decoupled-KL fix (Exp 8). Prior to Exp 8 the implementation used a single coupled KL constraint with a shared multiplier, which led to KL $\to 0$ (frozen policy) or KL $\to \infty$ (exploding critic) depending on initialisation. The fix implements separate α_μ/α_σ multipliers as specified in Abdolmaleki et al. [1]. All MPO experiments from Exp 8 onwards use the corrected implementation.

Network architecture. Both actor and critic share a 1D-CNN encoder over the camera observation lines (two convolutional layers, kernel size 5, channels $[16, 32]$, stride 2, embedding dimension 32) followed by fully connected layers (actor: 90 units; critic: 140 units). Hyperparameters are listed in Table 8.

Table 8: MPO hyperparameters (default configuration; all experiments unless noted).

Parameter	Symbol	Value
Batch size	B	256
Replay buffer	$\lvert\mathcal{B}\rvert$	50 000
Discount factor	γ	0.99
Target network	τ	0.005
Actor learning rate	η_π	4.5×10^{-4}
Critic learning rate	η_Q	1.0×10^{-3}
Dual learning rate	η_η	1.0×10^{-3}
KL samples (E-step Q)	N_Q	80
KL samples (M-step)	N_π	40
KL temperature constraint	ε_η	0.1
KL mean constraint	ε_μ	0.1
KL variance constraint	ε_σ	0.01
Warmup episodes	—	5
Training episodes	—	50 (except Exp 14: 350)
Eval episodes	—	2
Environment seed	—	7 (Exps 1–13)

2.10 Observation and action space

Observations. The controller observation is assembled at each controller tick from:

- **Attitude scalars:** body angle θ_{body} [rad], satellite body rate ω_{sat} [rad/s].
- **Orbit scalar:** true anomaly θ_{orbit} [rad].

- **Camera lines:** primary and secondary observation-line arrays (per-cell class labels: background, target identity, or cloud-blocked).
- **Mission context:** remaining shutter budget (dimensionless count); binary target-captured mask (one entry per target area); signed bearing error toward each target [rad].

Actions. Both SAC and MPO policies output a 2-dimensional continuous action $[a_\tau, a_s] \in [-1, 1]^2$:

- **Torque mode:** $a_\tau \mapsto \tau_{\text{cmd}} = a_\tau \cdot \tau_{\text{max}}$ [N m] (direct RW torque request).
- **Vector mode:** $a_\tau \mapsto f_n = 45^\circ \cdot a_\tau$ (nadir-relative pointing offset); the OBC PD controller converts this to a torque command.
- **Shutter:** $a_s \mapsto (a_s + 1)/2 \in [0, 1]$; boolean shutter command when mapped value > 0.5.

The torque versus vector mode is a configuration choice at training time; experiments from Exp 11 onward use vector mode as the default.

2.11 Simulation architecture

Training, evaluation, and visualisation share one simulated episode per roll-out. The reward, the evaluation KPIs, and the exported videos therefore reflect the same altitude, cloud layout, attitude trajectory, and camera observations. Mission randomisation is controlled by a fixed integer seed so that the deterministic baseline and each learned configuration are compared under identical environment draws.

3 Experiments

Fourteen experiments were conducted between 2026-06-28 and 2026-07-02 to map the reward and action-space design required for cloud-aware autonomous pointing. Each run tests one concrete design choice; the sequence is cumulative rather than a search for a single winning configuration. The arc divides into three phases: (i) diagnosing why naive training setups fail and whether observation structure helps (Exps 1–6), (ii) reward shaping and action-space design (Exps 4–9), and (iii) MPO-specific stabilisation and further ablations (Exps 8–12), followed by scale and cadence probes (Exps 13–14).

Experimental protocol. A fixed random seed (7) ensures identical altitude draws, cloud layouts, and target positions across all compared runs. All experiments share the simulation setup described in Section 2.11.

Before training begins, five *warmup episodes* pre-fill the replay buffer (Section 2.1) using the deterministic baseline policy under the same action interface and reward stack as the run that follows. This was not the initial choice: early attempts using random torque commands or a zero-torque coast strategy yielded no reward signal, because neither strategy ever issues capture commands that earn positive credit. The baseline policy, by contrast, reliably captures targets on clear passes. Even so, only around ten capture actions fall within a 516-step episode ($\approx$ 2% of all transitions), so the buffer is extremely sparse in rewarded transitions. The warmup data therefore serves a dual purpose: it bootstraps the replay buffer with at least some positive-credit transitions, and the mean warmup return is the matched *average baseline return* for that configuration.

Two *evaluation episodes* then run the learned policy without gradient updates; mean eval return is the primary post-training metric. A run is said to show a *learning signal* if its per-episode training return trends upward over training, indicating the policy has discovered the reward structure. Concretely, this requires at least one training episode to substantially exceed the average baseline return and for the trend to be sustained rather than a single outlier. Runs without a learning signal converge to a flat return plateau from episode 1 onward, regardless of the number of training episodes, and are reported as *not supported*. In the verdict column of Table 9, *supported* means the design choice under test produced the expected learning or behavioural change (for example selective shuttering or a stable upward return trend); it does not by itself claim mission-ready outperformance of the baseline. Results are archived as JSON files per experiment; key metrics are cited in each subsection below and summarised after the last experiment.

3.1 Exp 1: Shutter threshold ablation

Motivation. Preliminary runs showed the MPO agent emitting ∼968 shutter commands per episode, far exceeding the budget of ten and firing the shutter on nearly every controller step. The hypothesis was that raising the continuous action threshold (above which a shutter command is issued) from 0.5 to 0.9 would mechanically reduce this saturation and, combined with a 15 s credit window around each accepted capture, create a learnable reward gradient.

Config and result. Two configurations (shutter threshold 0.5 and threshold 0.9) were run for 7 training episodes at $\Delta t = 1.5$ s. Both produced an identical eval return of -101.6 with ∼516 shutter commands per episode. The policy's shutter action was saturated near its maximum value (mean ≈ 0.999) at both thresholds. It fired the shutter on almost every step regardless of threshold setting. Neither showed a learning signal; every shutter command produced zero useful capture credit.

Verdict. Not supported. Threshold tuning is not an effective lever while the policy saturates the action dimension. The credit window creates latent mid-episode reward bursts but does not fix credit assignment.

3.2 Exp 2: Modular encoder ablation

Motivation. The controller observation is 105-dimensional and includes two length-50 per-target arrays: bearing errors and capture-mask flags. Flattening those arrays into a single MLP may be sample-inefficient compared to embedding each array as a group [4]. Variant A1 replaces the flat concat with a Linear($50 \to 8$) embedding on each target array (camera lines unchanged).

Config and result. SAC sparse, 7 training episodes. A0 (flat): eval -78.9, 223 shutters/ep. A1 (structured encode): eval -86.5, 98 shutters/ep. Neither variant showed a learning signal; both produced zero meaningful captures. A1's critic loss diverged (3190 vs A0's 1420 at episode 7), indicating training instability in the structured branch.

Verdict. Not supported. Structured encoding of the per-target arrays shows no benefit in the 7-episode non-learnable slice. The same comparison is repeated under a longer protocol in Exp 6.

3.3 Exp 3: SAC vs. MPO algorithm comparison

Motivation. Exps 1–2 both used MPO; learning failure could be algorithm-specific. Exp 3 runs SAC (sparse reward) and MPO (dense reward) under matched protocol to isolate the algorithm–reward interaction [1, 5].

Config and result. Early-abort patience of 20 episodes. SAC sparse (early-abort at ep 37): eval -22.2, clear learning signal, best training return +5.3. MPO dense (early-abort at ep 22): eval -51.6, no learning signal, KL divergence $\approx 3.8 \times 10^5$. SAC clearly separated from MPO: the SAC policy converged while MPO's policy update blew up.

Verdict. Supported. SAC learns under sparse reward; MPO dense fails in this configuration. Algorithm choice matters; MPO's failure traced to an unconstrained M-step dual, motivating the fix in Exp 8.

3.4 Exp 4: Vector pointing mode

Motivation. Outputting a direct reaction-wheel torque conflates two problems: target scheduling and rate control. The vector pointing interface delegates rate control to a proportional–derivative loop and exposes a nadir-relative offset to the policy, reducing the action space the learner must shape for scheduling.

Config and result. SAC sparse, 50 training episodes. Ref0 (torque): eval −11.1, clear learning signal, best training return +8.4. Ref1 (vector): eval −24.1, clear learning signal, best training return +95.8. Ref1's eval was worse, but video of training episode 11 (+95.8) showed deliberate off-nadir scheduling and sparse mid-orbit shutter firings. This showed a qualitatively different and promising behaviour. End-of-episode shutter spam after budget exhaustion remained apparent in Ref1.

Verdict. Partial. Vector mode is valid and both variants learn, but the eval gap between them is unexplained. The shutter-budget ignorance motivates Exp 7. Vector pointing mode is retained as the preferred action interface for subsequent experiments.

3.5 Exp 5: MPO network width ablation

Motivation. Before investigating the MPO dual, the hypothesis that a 90/140-unit head was capacity-limited was tested with three widths: S (90/140), M (140/256), L (256/512). This was gated after the encoder experiment following de Bruin et al. [2] suggestions on structure-before-capacity.

Config and result. MPO sparse, 50 training episodes. All three widths produced identical eval −51.6 and identical KL freeze/explosion patterns. Post-episode-1 returns were flat; every shutter command produced zero useful capture credit across all three width variants.

Verdict. Not supported. Network width is not the bottleneck; the shared collapse pattern points to the learning algorithm.

3.6 Exp 6: Structured target-array encoding

Motivation. Exp 2 tested structured encoding of the per-target bearing and capture-mask arrays in a short non-learnable slice. Exp 6 repeats that single architectural delta under the longer SAC sparse protocol that Exp 3 showed can learn: flat concatenation (A0) versus a Linear(50 → 8) embedding of each target array before fusion (A1). Camera observation lines are unchanged; this is not image compression.

Config and result. SAC sparse, torque mode, 50 training episodes. Matched warmup baseline +393.2 on both arms. A0 (flat): best train +11.2, eval −3.9, learning signal present. A1 (structured encode): best train +137.2, eval +80.9, learning signal present. Both arms remain far below the matched baseline and still issue hundreds of shutter commands per eval episode.

Verdict. Supported. Given a learnable protocol, structured encoding of the per-target arrays beats flat concatenation on return. The Exp 2 null result was protocol-limited; selective budget-aware shuttering still required the later reward work (Exps 7 and 9).

3.7 Exp 7: SAC vector + budget-exhausted shutter penalty

Motivation. Exp 4 video (Ref1 ep 11) showed mid-orbit sparse shuttering but persistent shutter spam after the 10-capture budget was exhausted. The budget-exhausted penalty adds $-k_{\text{waste}}$ to the step reward whenever the agent commands a shutter with zero remaining budget (Section 2.7). Torque-path SAC with a static threshold (Exp 1) failed to stop spam; a direct negative reward on the offending action is the minimal correct lever.

Config and result. SAC sparse + vector + budget penalty, 50 training episodes. Eval +10.6 (vs Ref1 baseline −24.1, Δ = +34.7). Train shutter commands: 478/ep vs 7071/ep Ref1 (−93%). Train meaningful shutter fraction: 0.220 vs 0.015 Ref1 (15×). Post-peak return std: 17.5 vs 28.4 (−38%).

Verdict. **Supported.** Budget-exhausted penalty is an effective design lever: shutter spam drops sharply and selective scheduling emerges. This reward term is kept for all subsequent experiments.

3.8 Exp 8: MPO decoupled-KL dual fix (torque mode)

Motivation. Exps 3 and 5 both showed MPO freezing or exploding at KL $\approx 3.8 \times 10^5$ or ≈ 0. Investigation identified the root cause: the M-step used a single coupled Lagrange multiplier for both mean and variance KL constraints, instead of the two independent multipliers specified in Abdolmaleki et al. [1]. Exp 8 validates the corrected decoupled-KL dual in torque mode against the Exps 3/5 comparators.

Config and result. MPO (fixed dual) sparse torque, 50 training episodes, seed 7. Clear learning signal; eval −81.1 vs pre-fix floor −51.6. Best train +8.7 (ep 28). KL converged to 0.021; temperature dual settled at 0.788. Train torque saturation fraction: 0.951 (vs $\approx$ 0.998 pre-fix). No meaningful eval captures.

Verdict. **Partial.** The algorithm fix is confirmed: bounded KL and rising train returns. Eval return remains below the matched average baseline return; torque mode still saturates and safe-mode activates frequently, motivating Exps 10 and 11.

3.9 Exp 9: SAC shutter reward split (waste penalty off)

Motivation. Exp 7 had both the budget-exhausted penalty and the within-budget waste penalty active. Since applied-capture credit already scales with image quality and cloud fraction (Eq. 16), a second explicit waste penalty on the same action may over-constrain shutter exploration. Exp 9 tests turning off the waste penalty while keeping the budget-exhausted term.

Config and result. SAC sparse + vector + budget penalty (W off), 50 training episodes. Eval +106.4 (vs Exp 7 +10.6, Δ = +95.8). Best train +169.9 (ep 36). Train meaningful fraction: 0.286 vs 0.220 (Exp 7). Eval meaningful fraction: 0.667 vs 0.400. Post-budget shutter commands: 75 (penalty still active).

Verdict. **Supported.** Relaxing the over-constraining within-budget waste penalty yields a large eval improvement. This is the best SAC configuration in the series and defines the reward stack used as the reference for later MPO comparisons.

3.10 Exp 10: MPO safe-mode penalty

Motivation. Exp 8 showed safe-mode activating 5–6 times per episode in torque mode with no reward signal. A per-step penalty during safe-mode recovery intervals (k = 2.0) was hypothesised to reduce activations and improve applied torque credit assignment.

Config and result. MPO (fixed dual) sparse torque + safe-mode penalty ($k = 2.0$), 50 training episodes. Eval −510.4 (vs Exp 8 −81.1, catastrophic regression). Training return improved from −1168 (ep 0) to ≈ −350 (ep 23) but stagnated. Train shutter commands dropped from 153 (ep 0) to 2–7 (eps 30–49).

Verdict. Not supported. The penalty taught the agent to avoid all aggressive torque (including captures) rather than only avoiding safe-mode entry. Safe-mode activations co-occur with off-nadir pointing required for capture; the penalty suppressed both. Per-step safe-mode penalty at this coefficient is ruled out for torque-mode MPO. Vector mode (Exp 11) provides a structurally cleaner solution.

3.11 Exp 11: MPO fixed dual, vector mode

Motivation. Exp 10 ruled out reward-based safe-mode mitigation in torque mode. Exp 8 confirmed the dual fix; the natural next step is MPO in vector pointing mode, mirroring the Exp 4 SAC comparison but now with a working MPO agent. Vector mode avoids the safe-mode interference seen in torque mode.

Config and result. MPO (fixed dual) sparse vector, 50 training episodes, seed 7. Eval +45.5 (vs Exp 8 −81.1, $\Delta = +126.6$). Best train +82.3 (ep 42). Positive returns from episode 7 onward; monotonic improvement from ep 5. KL: 0.014; η: 0.014 (vs 0.788 in torque mode). Shutter commands: 148 (ep 0) → 8–10 (eps 30–49). The policy learned selective shuttering.

Verdict. Supported. Vector mode resolves the torque-mode MPO stagnation: the first MPO configuration with positive eval return. The temperature dual settles much lower in vector mode, reflecting a tighter value distribution in the simplified action space. Vector pointing mode is retained for subsequent MPO experiments. The remaining gap to SAC Exp 9 (+106.4) suggests the MPO branch has not yet been tested with the same reward stack as Exp 9.

3.12 Exp 12: MPO vector + torque-effort penalty ablation

Motivation. The training workflow disables the torque-effort regulariser in vector mode by default. Exp 12 re-enables effort ($k = 0.1$) while keeping all other Exp 11 settings.

Config and result. MPO (fixed dual) sparse vector + torque effort ($k = 0.1$), 50 training episodes. Eval +25.3 (vs Exp 11 +45.5, $\Delta = -20.2$). Best train +75.6 (ep 20). Four consecutive zero-return episodes at eps 36, 41, 42, 43. KL: 0.013; η: 0.027.

Verdict. Not supported. Torque effort in vector mode hurts performance: the effort penalty targets the pointing-controller wheel command, not the policy's pointing offset, creating a conflicting gradient that discourages off-nadir pointing irrespective of imaging context. Effort regularisation should remain disabled in vector mode.

3.13 Exp 13: MPO learning-cadence sweep

Motivation. The MPO agent performs one gradient update per environment step by default. If the agent learns faster than the environment provides signal, or if gradient noise from a partially filled replay buffer impedes early learning, a different cadence may help. Exp 13 evaluates three cadence profiles: update every step, every 10 steps, and every 50 steps.

Config and result (partial). MPO (fixed dual) torque sparse, 10 training episodes per configuration (shorter than the 50-episode vector runs). Average baseline return (5 warmup episodes): +410.2. Update every step (control): eval −38.6, best train −22.9. Update every 10 steps: eval −46.4, best train −38.9.

Update every 50 steps: eval −15.4, best train −31.4 (best eval in the short sweep; still below Exp 11 +45.5).

Verdict. **Partial.** Sparser gradient updates (every 50 steps) outperform the per-step and every-10-steps configurations in this short slice, but results remain well below the later vector-mode MPO configuration (Exp 11). A longer training slice is needed for a definitive cadence conclusion.

3.14 Exp 14: MPO multi-environment target selection

Motivation. All prior experiments use a single environment seed (7) with a fixed target corridor. Generalisation to diverse orbital configurations, altitudes, and cloud draws requires training across multiple environment draws. Exp 14 attempted a 350-episode multi-environment MPO run to test this scaling hypothesis.

Config and result. MPO (fixed dual) vector sparse, 350 training episodes, multi-seed draws. The run aborted partway through training due to an infrastructure error during the long multi-environment run. Partial mid-training metrics were available (mean score 0.228) but are not directly comparable to the episodic eval return metric used in the rest of the series.

Verdict. **Not supported** (run failure; results inconclusive). Multi-environment training remains an open direction. Recommended follow-up: increase episode budget and match the evaluation metric to the rest of the series.

3.15 Summary comparison

Reward modes. Table 9 groups configurations by which reward terms are active (Section 2.7):

- **Sparse:** capture credit only when a shutter yields a first-time, in-budget observation of a target (Eq. 16); most steps carry zero capture reward.
- **Dense:** Exp 3 MPO configuration; adds dense shaping on top of capture events, tested against sparse SAC under matched protocol.
- **Budget penalty:** a negative term when the agent commands a shutter after the ten-capture budget is exhausted; Exp 7 uses this together with the within-budget waste penalty.
- **Waste penalty:** a negative term on accepted shutters with near-zero applied capture credit; Exp 9 removes it while keeping the budget-exhausted term.
- **Safe-mode penalty:** a per-step penalty while attitude safety recovery is active (Exp 10).
- **Effort:** a quadratic penalty on commanded torque (Exp 12 in vector mode).

The *Baseline* column is the mean return of the five warmup episodes under that row's action interface and reward mode. *Eval* is the mean of the two evaluation episodes after training. Verdicts use the definition in Section 3.

4 Results

This section summarises what the experiment sequence revealed about reward shaping and action-space design. Eval return and shutter behaviour are interpreted against each run's matched average baseline return (five warmup episodes under the same action interface and reward mode; Table 9). A positive Δ_{base} means the learned policy beat that matched baseline on the fixed evaluation seed; it does not by itself mean the training problem is solved in general.

Table 9: Experiment comparison. Algorithm is SAC or MPO (star: decoupled-KL dual fix from Exp 8). Action: torque = direct reaction-wheel command; vector = nadir-relative pointing offset. Protocol: five warmup episodes and two eval episodes for every row; train length varies.

Exp	Change	Alg	Action	Reward mode	Baseline	Best train	Eval	Verdict
1	Shutter threshold 0.5	MPO	Torque	Sparse	+393.2	−99.9	−101.6	Not supported
1	Shutter threshold 0.9	MPO	Torque	Sparse	+393.2	−99.9	−101.6	Not supported
2	Flat target arrays	SAC	Torque	Sparse	+393.2	−72.4	−78.9	Not supported
2	Structured target encode	SAC	Torque	Sparse	+393.2	−75.1	−86.5	Not supported
3	Algorithm compare	SAC	Torque	Sparse	+393.2	+5.3	−22.2	Supported
3	Algorithm compare	MPO	Torque	Dense	+393.2	−51.6	−51.6	Not supported
4	Torque interface	SAC	Torque	Sparse	+393.2	+8.4	−11.1	Partial
4	Vector interface	SAC	Vector	Sparse	−48.5	+95.8	−24.1	Partial
5	Network width (all)	MPO	Torque	Sparse	+393.2	−51.6	−51.6	Not supported
6	Flat target arrays	SAC	Torque	Sparse	+393.2	+11.2	−3.9	Supported‡
6	Structured target encode	SAC	Torque	Sparse	+393.2	+137.2	+80.9	Supported‡
7	Budget penalty	SAC	Vector	Sparse + budget	−38.6	+124.6	+10.6	Supported
8	Decoupled dual	MPO⋆	Torque	Sparse	+393.2	+8.7	−81.1	Partial
9	Waste penalty off	SAC	Vector	Sparse + budget, waste off	−38.6	+169.9	+106.4	Supported
10	Safe-mode penalty	MPO⋆	Torque	Sparse + safe-mode	+410.2	−354.4	−510.4	Not supported
11	Vector after dual fix	MPO⋆	Vector	Sparse	+2.4	+82.3	+45.5	Supported
12	Effort in vector	MPO⋆	Vector	Sparse + effort	+2.4	+75.6	+25.3	Not supported
13	Cadence every step	MPO⋆	Torque	Sparse	+410.2	−22.9	−38.6	Partial
13	Cadence every 10 steps	MPO⋆	Torque	Sparse	+410.2	−38.9	−46.4	Not supported
13	Cadence every 50 steps	MPO⋆	Torque	Sparse	+410.2	−31.4	−15.4	Partial
14	Multi-environment	MPO⋆	Vector	Sparse	—	+87.6	n/a†	Not supported

Baseline = mean warmup return (5 episodes). Eval = mean of 2 episodes.

Exp 3 early-abort (patience 20). Exp 9 keeps budget penalty, disables within-budget waste.

‡ Exp 6: Supported for structured vs flat target-array encoding return gap; not a baseline win.

† Run aborted (infrastructure error during long multi-environment run).

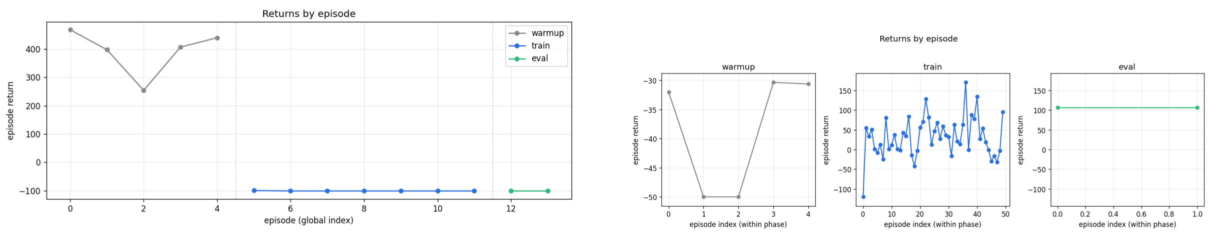
4.1 Master scoreboard

Table 9 is the master scoreboard: eval return, matched baseline return, and Δ_{base} for every closed experiment. The subsections below add cross-cuts (learning curves, shutter efficiency, safe-mode overhead, and SAC vs. MPO) rather than re-listing those numbers.

4.2 Learning progression

Episode-level learning curves mark the milestones on the scoreboard (Table 9):

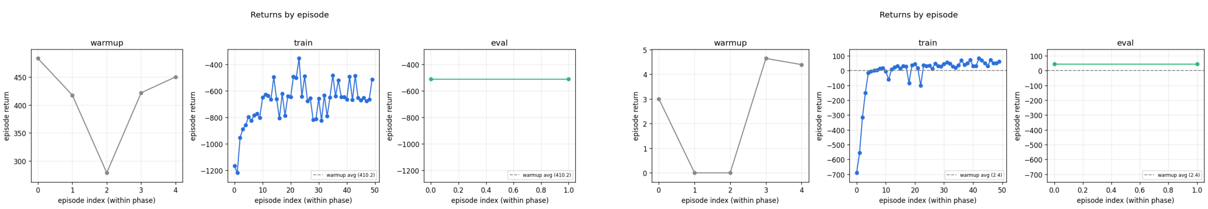
- **Exps 1–5:** Threshold tuning, encoder variants, and algorithm/width ablations stay flat or degrade; no run in this phase reaches positive eval return.
- **Exp 3:** SAC with sparse reward is the first configuration with a rising training trajectory.
- **Exp 6:** Structured target-array encoding (A1) improves returns under the longer SAC sparse protocol (eval +80.9). Supported for the A1 vs. A0 encoder gap, not a baseline win: both arms remain below the matched warmup baseline and still spam the shutter.
- **Exp 7:** Budget-exhausted shutter penalty on vector-mode SAC yields the first budget-aware, selective positive eval return (+10.6) and cuts train shutter commands from ~7000 (Exp 4) to 478.
- **Exp 9:** Waste-penalty removal further improves eval return (+106.4); on eval, two thirds of capture attempts are scientifically valid (non-cloud, within budget, previously uncaptured target).
- **Exp 11:** After the Exp 8 dual fix, vector-mode MPO reaches +45.5, the first positive MPO eval return.



(a) Exp 1 (not learning)

(b) Exp 9 (learning)

Figure 6: Episode-return curves (warmup / train / eval). Exp 1 shutter-threshold MPO collapses to near-floor train and eval returns after a strong warmup; Exp 9 SAC shutter-split reaches positive eval returns (best supported SAC configuration).



(a) Exp 10 (not learning)

(b) Exp 11 (learning)

Figure 7: Episode-return curves for the MPO branch after the Exp 8 dual fix: Exp 10 safe-mode penalty stays deeply negative on train/eval, while Exp 11 vector-mode MPO recovers during train and finishes with positive eval return.

4.3 Shutter spam and budget efficiency

Shutter command count per episode tracks whether a policy has learned selective scheduling rather than firing continuously. A well-trained policy would issue at most ten commands per episode (one per target within budget), timed to clear-sky windows. The deterministic baseline issues exactly ten commands but ignores cloud content, so an estimated 30–40% of its captures are wasted on cloud-blocked or empty targets.

Table 10 tracks the spam reduction path.

Table 10: Shutter spam reduction across the SAC vector experiment chain. All figures are training means unless noted.

Experiment	Cmds/ep	Meaningful frac	Notes
Pre-study MPO (threshold 0.5)	968	0	Near-continuous shuttering
Exp 1 (thresholds 0.5 / 0.9)	516	0	Threshold ineffective
Exp 4 (SAC vector)	7071	0.015	Budget ignorance
Exp 7 (budget penalty)	478	0.220	−93% cmds vs Exp 4
Exp 9 (waste off)	—	0.286 / 0.667 (eval)	Best scheduling
Exp 11 (MPO vector)	8–10	—	Budget-aware MPO
Baseline (reference)	10	~0.60–0.70	Deterministic, cloud-blind

The budget-exhausted penalty (Exp 7) was the single most impactful reward change in the series: a 93% reduction in shutter commands with a simultaneous increase in eval return. Removing the waste penalty (Exp 9) added a further behavioural improvement on top of that configuration. Together, these results show that reward structure, not algorithm choice alone, determines whether selective shuttering emerges.

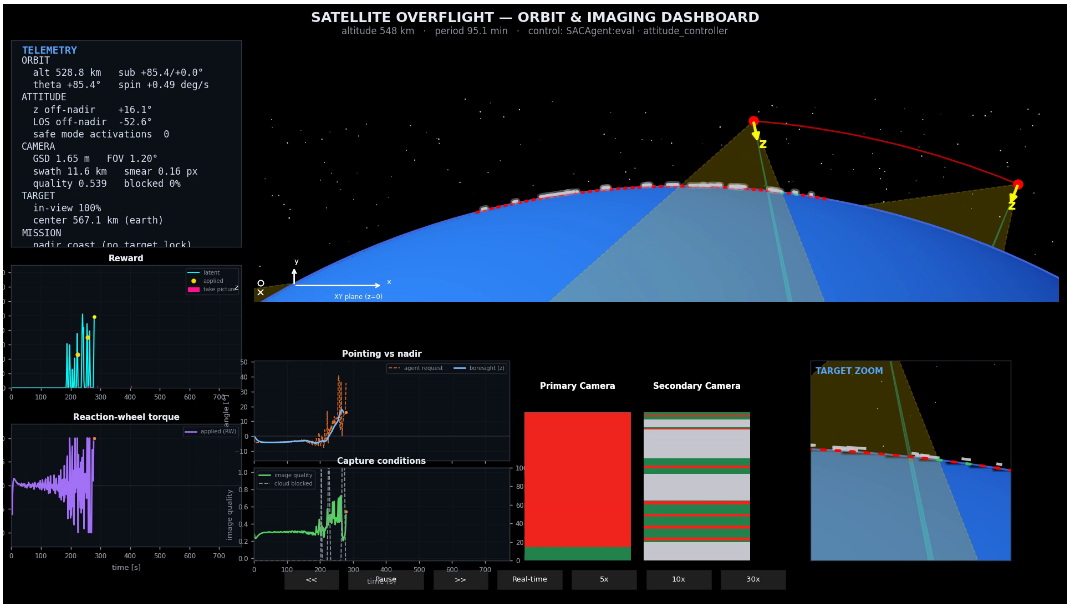


Figure 8: Eval episode from Exp 9 (SAC shutter-split) on the fixed evaluation seed. Reward panel shows a sparse cluster of applied rewards and take-picture markers during the engagement window rather than continuous shutter spam.

4.4 Safe-mode and safety overhead

In torque mode (Exps 1–5, 8) the safety controller activates 5–6 times per episode, locking out the agent for 60 s per event (Section 2.6). Each lockout consumes ≈40 controller steps with no capture opportunity. With 516 steps per episode and 5 lockouts, approximately 40% of the episode is unavailable to the agent.

Vector mode (Exps 4 Ref1, 7, 9, 11, 12) eliminates safe-mode lockouts in the experiment runs: the hold-last semantics prevent the pointing reference from exceeding θ_{hard}, so the safety controller is never triggered.

This structural difference explains much of the torque-versus-vector gap. The safe-mode penalty (Exp 10) attempted to replicate this improvement via reward shaping but produced the worst result in the series (−510.4), confirming that action-space redesign (vector mode) is preferable to reward penalties for aligning safety and learning.

4.5 Algorithm comparison: SAC vs. MPO

At matched protocol (sparse, vector, 50 episodes, seed 7), the best SAC configuration (Exp 9, +106.4) exceeds the best MPO configuration (Exp 11, +45.5) on eval return by 60.9 points. The gap is partially explained by reward configuration: Exp 9 uses the waste-off, budget-on reward while Exp 11 uses the budget-on, waste-on default. An MPO run with the Exp 9 reward settings was not completed within the project timeline.

Both best configurations exceed their matched average baseline return on this seed (Exp 9: +106.4 vs −38.6; Exp 11: +45.5 vs +2.4). That is encouraging for the design search but not sufficient evidence of robust mission-level superiority: evaluation uses only two episodes at a fixed seed, and several intermediate configurations never learned at all. The more durable finding is behavioural. Both algorithms can learn selective shuttering once reward and action-space choices are correct. That is a stronger claim than reliable baseline outperformance under all training conditions.

5 Discussion

5.1 What the experiment sequence taught

The motivating idea, use onboard vision to shutter only on clear targets, is simple to state but hard to train. The fourteen experiments were not a single attempt to beat the baseline; they were a structured search for reward and action-space configurations that make learning possible. Three lessons emerged. Per-experiment returns and shutter counts are reported in Sections 3 and 4; this subsection interprets why the levers mattered.

1. Reward structure before algorithm. Threshold tuning, network width, and algorithm choice alone did not unlock selective shuttering. Structured encoding of the per-target arrays (Exp 6) improved returns once a learnable protocol existed, but both encoder arms still spammed the shutter and stayed below the matched baseline. Selective, budget-aware behaviour emerged only after the reward told the agent what mattered at the action clock: sparse capture credit, a budget-exhausted penalty that stops shutter spam, and removal of a redundant within-budget waste term. Reward engineering remained the dominant lever for that behaviour.

2. Action-space design before reward fine-tuning. Delegating rate control to the onboard pointing controller (vector mode) is structurally important relative to raw torque commands. Vector mode removes safe-mode lockouts in the experiment runs (Section 4.4), resolving a credit-assignment problem that no reward penalty could fix cleanly. This is a design result: the agent should command a pointing offset, not raw wheel torque, for cloud-aware scheduling in this environment (Section 5.3).

3. Algorithm stability as a prerequisite. An unconstrained MPO trust-region update blocked the MPO branch until the dual-variable bug was corrected. After that fix, vector-mode MPO learned without a further reward redesign. Implementation correctness can masquerade as algorithm failure; that check comes before comparing reward or action-space variants.

5.2 Answer to the research question

Cloud-aware autonomous scheduling is not trivial to train in this simulation. A minimal viable training stack requires at least:

- a sparse capture-credit reward with a budget-exhausted shutter penalty;
- vector pointing mode (nadir-relative offset) rather than direct torque commands;
- a correctly implemented MPO trust-region update, or SAC as the more forgiving default;
- baseline-policy warmup to seed the replay buffer with rare positive-credit transitions.

With that stack, both SAC and MPO learn selective shuttering absent in the deterministic baseline. Several configurations that looked reasonable (threshold tuning, network width, safe-mode penalties, torque effort in vector mode) failed to produce learning or made it worse. Eval return on seed 7 exceeded the matched average baseline return for the best SAC and MPO configurations, but the evaluation protocol is too narrow to claim robust mission superiority; the durable output of the semester is the mapped design space and the list of rejected shortcuts.

5.3 Control hierarchy and action interface

Development did not begin with the interface the experiments ultimately recommend.

From torque to safety arbitration. The first experimental interface exposed raw reaction-wheel torque to the learning agent. Unconstrained policies could command aggressive slews; in simulation this produced sustained rotation that would be unacceptable on flight hardware. Review with system engineers motivated treating attitude safety as non-negotiable: every agent command, whether torque or pointing request, passes through a safety controller that can taper, reject, or replace commands with a safe-mode recovery sequence (Section 2.6). That stack is shared by the deterministic baseline and all learned policies.

Action modes tested. Three action interfaces were explored:

- **Torque mode:** the policy commands wheel torque directly (Exps 1–6, 8, 10, 13).
- **Vector mode:** the policy commands a nadir-relative pointing offset; the OBC converts it to PD torque (Exps 4, 7, 9, 11, 12, 14).
- **Target selection:** a higher-level discrete choice of which target to engage (Exp 14). Started as a hypothesised next simplification after torque and vector; the run did not complete, so whether this interface is easier to train remains open.

Segregation of control layers. The experiment arc supports separating what the agent should learn from what the OBC should execute. Learning precise torque profiles or continuous pointing offsets is not the primary scheduling problem in this environment: the agent must decide *when* to expose the shutter given cloud feedback, while rate control and maneuver execution belong downstream. Vector mode reflects that split and improved learning in practice (Exps 4, 11). A torque-trained agent might in principle discover rich low-level strategies, for example multiple captures within one forward-motion-compensation maneuver. It is a further *hypothesis*, not a result of this report, that raising the action interface to explicit target selection (plus shutter timing) would simplify credit assignment further; Exp 14 was begun on that conviction but time ran out before it could be validated, and implementation or gradient issues in that branch have not been ruled out. Energy-optimal wheel commands remain a distinct optimisation problem; they should be trained or tuned in a separate low-level loop rather than entangled with cloud-aware target scheduling.

Intended end state (working hypothesis). A plausible long-term stack is three levels: mission intent from the ground or onboard planner, a vision-driven scheduling policy (target and shutter decisions), and OBC execution with PD pointing and hard safety limits. Vector mode is the highest-level interface validated here; target selection is the candidate next step named above. This semester maps the scheduling layer under vector actions in a simplified 2D simulator; the lowest layer is modelled but not co-trained for efficiency.

5.4 Rejected paths

Table 11 summarises design choices that were tested and ruled out.

Table 11: Rejected experimental directions with rationale.

Experiment	Design choice tested	Why rejected
—	MPC pointing controller	Superseded by onboard PD pointing; lower scope and comparable results
—	Latent-only dense reward	Dense reward tested at 10× applied scale; did not unblock MPO
Exp 1	Shutter threshold 0.5 → 0.9	Policy saturates shutter output regardless of threshold
—	15 s capture credit window	Latent bursts visible; applied capture remains zero
Exp 5	MPO network width ablation	Identical collapse at all widths; capacity not the bottleneck
Exp 9	Within-budget waste penalty (on)	Redundant with apply-credit sparsity; penalising same failure mode twice
Exp 10	Safe-mode penalty	Correlation with aggressive pointing suppresses useful captures
Exp 12	Torque effort in vector mode	Pointing-controller torque ≠ policy action; conflicting gradient

5.5 Limitations

Two-dimensional simulation. The entire dynamics model, camera model, and reward signal operate in a single orbit plane. Real three-axis attitude control involves cross-coupling between roll, pitch, and yaw; disturbance torques from solar pressure, magnetic field, and gravity gradient; and three-dimensional off-nadir imaging geometry. The 2D model provides a tractable learning environment but the policies are not directly transferable to hardware.

Single-pass, single-orbit mission. Each episode models one overflight of a fixed 50-target corridor. Revisit scheduling, downlink windows, and multi-orbit memory are not represented. The learned policy has no access to observations from prior passes.

Simulation-only evaluation. No hardware-in-the-loop or software-in-the-loop tests were conducted. The camera model is a geometric ray tracer with an analytic smear formula; radiometric calibration, MTF, and sensor noise are absent. Generalisation from this environment to a real EO instrument is an open question.

Fixed-seed evaluation. All primary eval returns are computed on two episodes at seed 7. The train-to-eval gap visible in several experiments (e.g. Exp 9 train +169.9 vs eval +106.4) may partially reflect evaluation seed dependence. A robust evaluation across ≥10 seeds is recommended before citing the best numbers as reliable performance estimates.

Budget and episode length. The 10-capture budget over 516 steps is an engineered constraint that simplifies mission planning but may not represent realistic EO tasking where capture rate, data volume, and ground station contact are coupled objectives.

5.6 Deferred extensions

The following directions are documented as promising but were not completed within the project:

- **MPO with Exp 9 reward config.** Exp 11 reached +45.5 without the waste-penalty-off configuration used in Exp 9. A direct MPO run at Exp 9 reward settings would quantify the algorithm gap more cleanly.
- **Target-selection interface (Exp 14 rerun).** Hypothesis: discrete target choice plus shutter timing delegates pointing entirely to the OBC and may simplify learning relative to vector mode. Exp 14 began testing this idea but crashed before a conclusive result; a rerun must also rule out implementation or gradient bugs in that branch, not only extend training time.
- **Three-axis attitude extension.** Moving from 2D to 3D dynamics is the prerequisite for any hardware-facing transfer. The simulation architecture supports this extension via replacement of the single-axis RW model with a 3-wheel assembly and attitude quaternion propagation.
- **Real imagery and radiometric reward.** Replacing the geometric smear quality metric with a model-based image quality score derived from sensor noise and scene contrast would couple the reward more directly to downstream science value.

6 Conclusion

We set out to answer a question that looks simple from the outside: if cloud-aware autonomous scheduling is obviously better than a fixed capture plan, what reward shaping and action-space design are needed to train such a policy in simulation? The semester work shows that the answer is not trivial.

A reproducible simulation and training platform supported a systematic search over that design space. Three levers dominated: reward information at the action clock, vector pointing rather than raw torque, and a correctly implemented MPO trust-region update. With that stack, both SAC and MPO developed selective shuttering absent in the deterministic baseline. Reliable quantitative outperformance of the baseline on mission return was not established; the main contribution is the mapped design space and the rejected shortcuts (Section 5).

Outlook. Near-term work centers on broader evaluation seeds, longer training runs, and higher-level control authority. Promising but unfinished extensions are documented in Section 5.6 and are not restated here. Those items remain outside the semester claim.

Acknowledgements

I thank Prof. Dr. Marco Hutter (ETH RSL) for academic supervision, Alejandro Torrents Rufas and Mathias Burkhalter (Beyond Gravity) for industry supervision, and Beyond Gravity for hosting this semester project.

A Supplementary material

A.1 Repository

The full source code, experiment scripts, and run artifacts are available at:

`https://github.com/cgriede/autonomous-satellite-control-eth-sem-proj`

A.2 Source index

Table 12 lists the high-level entry points for the components described in this report. All paths are relative to the repository root.

Table 12: High-level source index.

Component	Entry point
Earth & geodetic constants	`backend/environment_definition/constants/EARTH.py`
Mission profile (altitude, targets)	`backend/environment_definition/constants/MISSION.py`
Satellite inertia & camera optics	`backend/environment_definition/constants/SATELLITE.py`
Attitude safety thresholds	`backend/environment_definition/constants/ATTITUDE_SAFETY.py`
Simulation timing	`backend/environment_definition/constants/SIMULATION.py`
Reward weights	`backend/environment_definition/constants/AUTONOMOUS_CONTROL_REWARD.py`
Canonical simulation entry point	`backend/simulation/run_simulation.py`
Camera image quality model	`backend/simulation/image_quality.py`
Attitude safety controller	`backend/simulation/attitude_controller.py`
Deterministic baseline policy	`backend/s01_utils/baseline_overflight.py`
RL agent (MPO / SAC)	`backend/autonomous_control/controller_agent.py`
Training workflow	`backend/autonomous_control/training_workflow.py`
Experiment scripts	`backend/scripts/experiments/`
Per-run return KPIs (warmup/train/eval)	`backend/autonomous_control/runs/<run-id>/summary_metrics.json`

A.3 Live parameter record

Hyperparameters and constants are maintained in:

- `docs/presentation/environment-hyperparameters.md`
- `docs/presentation/machine-learning.md`
- `docs/presentation/technical-constants.md`
- `docs/presentation/physical-reference.md`

A.4 Artifact paths

- Training outputs: `backend/autonomous_control/runs/<run-id>/`
- Experiment results: `backend/scripts/experiments/<slug>/results/`

References

[1] Abbas Abdolmaleki, Jost Tobias Springenberg, Jonas Degrave, Steven Bohez, Yuval Tassa, Daniele Belov, Nicolas Heess, and Martin Riedmiller. Maximum a posteriori policy optimisation. *arXiv preprint arXiv:1806.06920*, 2018.

[2] Tim de Bruin, Jens Kober, Karl Tuyls, and Robert Babuska. Integrating state representation learning into deep reinforcement learning. *IEEE Robotics and Automation Letters*, 3(3):1394–1401, 2018.

[3] Benedetta Ferrari, Jean-François Cordeau, Maxence Delorme, Manuel Iori, and Roberto Orosei. Satellite scheduling problems: A survey of applications in Earth and outer space observation. *Computers & Operations Research*, 171:106875, 2024. doi: 10.1016/j.cor.2024.106875.

[4] Yury Gorishniy, Ivan Rubachev, Valentin Khrulkov, and Artem Babenko. On embeddings for numerical features in tabular deep learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[5] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, 2018.

Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Declaration of originality

The signed declaration of originality is a component of every written paper or thesis authored during the course of studies. In consultation with the supervisor, one of the following three options must be selected:

- ○ I confirm that I authored the work in question independently and in my own words, i.e. that no one helped me to author it. Suggestions from the supervisor regarding language and content are excepted. I used no generative artificial intelligence technologies[1].
- ○ I confirm that I authored the work in question independently and in my own words, i.e. that no one helped me to author it. Suggestions from the supervisor regarding language and content are excepted. I used and cited generative artificial intelligence technologies[2].
- ◉ I confirm that I authored the work in question independently and in my own words, i.e. that no one helped me to author it. Suggestions from the supervisor regarding language and content are excepted. I used generative artificial intelligence technologies[3]. In consultation with the supervisor, I did not cite them.

Title of paper or thesis:

Cloud-Aware Learning-Based Attitude Control for Rainforest Earth Observation

Authored by:
If the work was compiled in a group, the names of all authors are required.

Last name(s):	First name(s):
Grieder	Cédric Sebastien

With my signature I confirm the following:
- I have adhered to the rules set out in the Citation Guide.
- I have documented all methods, data and processes truthfully and fully.
- I have mentioned all persons who were significant facilitators of the work.

I am aware that the work may be screened electronically for originality.

Place, date	Signature(s)
Gili Air, Indonesia, 21.07.2026	Grieder

If the work was compiled in a group, the names of all authors are required. Through their signatures they vouch jointly for the entire content of the written work.

[1] E.g. ChatGPT, DALL E 2, Google Bard
[2] E.g. ChatGPT, DALL E 2, Google Bard
[3] E.g. ChatGPT, DALL E 2, Google Bard